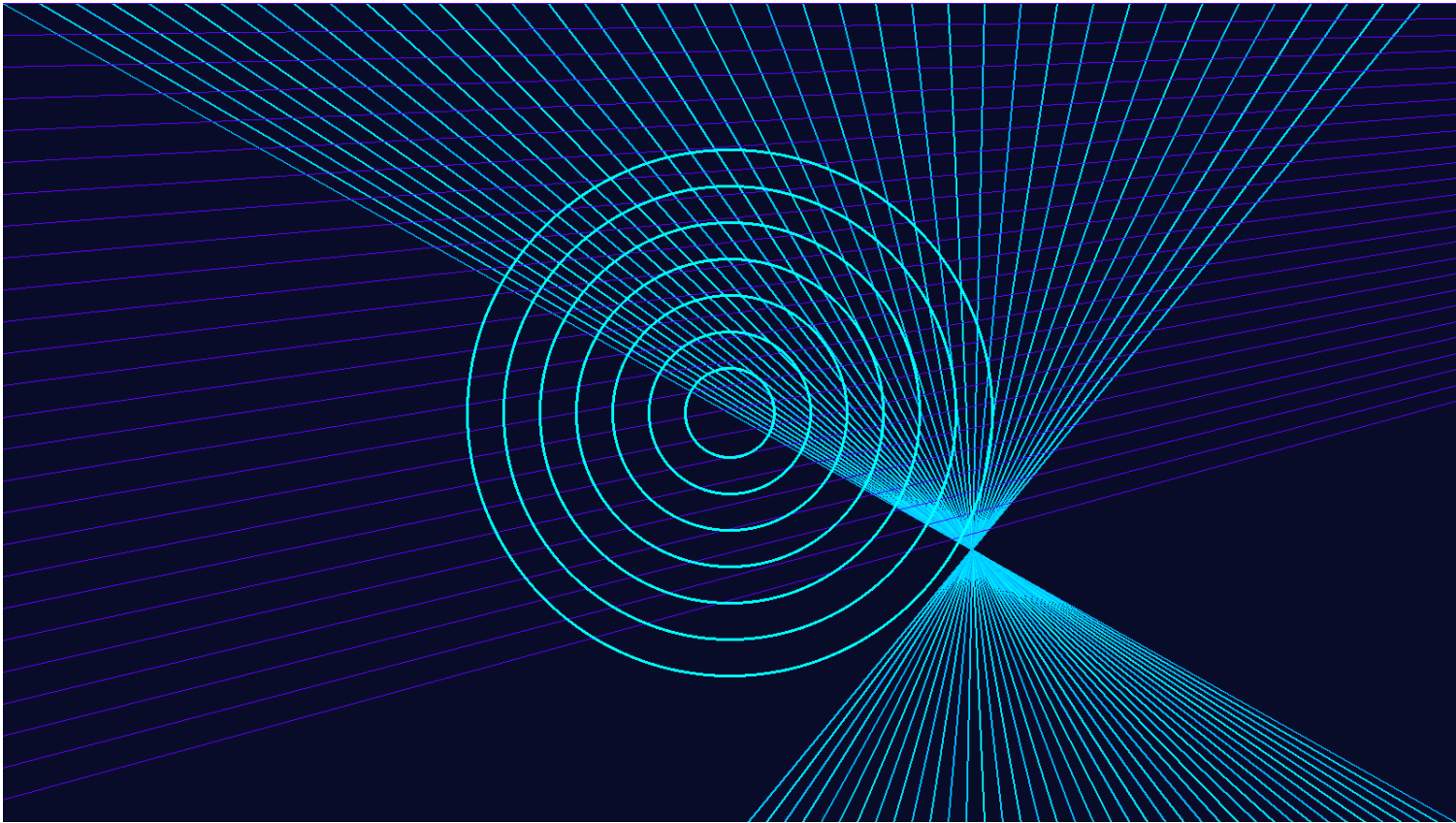




MODUL PRAKTIKUM

SISTEM TERDISTRIBUSI

Pertemuan : Praktikum 10

Dosen Pengampu : Ali Nur Ikhsan, M.Kom.

Program Studi Informatika

Fakultas Ilmu Komputer

Universitas Amikom Purwokerto

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Autentikasi & Otorisasi: JWT

Authentication di ASP.Net Core Web API

TUJUAN PRAKTIKUM

Setelah menyelesaikan praktikum ini mahasiswa mampu:

1. Mahasiswa memahami konsep autentikasi dan otorisasi.
2. Mahasiswa mampu membuat endpoint login.
3. Mahasiswa mampu menghasilkan token JWT.
4. Mahasiswa mampu mengamankan endpoint Web API menggunakan [Authorize].
5. Mahasiswa mampu menguji JWT menggunakan Swagger atau Postman.

Langkah Praktikum:

1. Buka Project Berikut: https://drive.google.com/drive/folders/1ijfkhf_CUj-8DLdeZUII_9VHHgwm6lmW?usp=sharing
2. Tambahkan Package Manager Console:
Install-Package Microsoft.AspNetCore.Authentication.JwtBearer -Version 8.0.0
3. Buat class baru di folder Models - LoginModel.cs. Lalu ketik kode program berikut.

```
namespace WebAPI_Akademik.Models
{
    1 reference
    public class LoginModel
    {
        2 references
        public string? Username { get; set; }
        1 reference
        public string? Password { get; set; }
    }
}
```

Mengatur JWT di Program.cs

Tambahkan namespace Berikut

```
using Microsoft.EntityFrameworkCore;
using System.Text.Json.Serialization;
using WebAPI_Akademik.Models;
//tambahkan namespace berikut
using Microsoft.AspNetCore.Authentication.JwtBearer;
using Microsoft.IdentityModel.Tokens;
using System.Text;
```

Tambahkan kodingan Berikut:

```

builder.Services.AddControllers()
    .AddJsonOptions(options =>
    {
        options.JsonSerializerOptions.ReferenceHandler = ReferenceHandler.IgnoreCycles;
    });

builder.Services.AddDbContext<AppDbContext>(options =>
    options.UseSqlServer(builder.Configuration.GetConnectionString("DefaultConnection")));

//tambahkan kode berikut
var key = "kunci_rahasia_jwt_minimal_32_karakter";

builder.Services.AddAuthentication(JwtBearerDefaults.AuthenticationScheme)
    .AddJwtBearer(options =>
    {
        options.TokenValidationParameters = new TokenValidationParameters
        {
            ValidateIssuer = false,
            ValidateAudience = false,
            ValidateIssuerSigningKey = true,
            IssuerSigningKey = new SymmetricSecurityKey(Encoding.UTF8.GetBytes(key))
        };
    });

//tambahkan kode berikut
app.UseAuthentication();

app.UseAuthorization();

app.MapControllers();

app.Run();

```

4. Buat Controller – AuthController – ketikkan kode berikut:

```

namespace WebAPI_Akademik.Controllers
{
    [ApiController]
    [Route("api/[controller]")]
    0 references
    public class AuthController : ControllerBase
    {
        private readonly string key = "kunci_rahasia_jwt_minimal_32_karakter";

        [HttpPost("login")]
        0 references
        public IActionResult Login(LoginModel login)
        {
            if (login.Username == "admin" && login.Password == "admin123")
            {
                var claims = new[]
                {
                    new Claim(ClaimTypes.Name, login.Username)
                };

                var securityKey = new SymmetricSecurityKey(Encoding.UTF8.GetBytes(key));
                var credentials = new SigningCredentials(securityKey, SecurityAlgorithms.HmacSha256);

                var token = new JwtSecurityToken(
                    claims: claims,
                    expires: DateTime.Now.AddHours(1),

                    var credentials = new SigningCredentials(securityKey, SecurityAlgorithms.HmacSha256);

                    var token = new JwtSecurityToken(
                        claims: claims,
                        expires: DateTime.Now.AddHours(1),
                        signingCredentials: credentials
                    );

                    var tokenString = new JwtSecurityTokenHandler().WriteToken(token);

                    return Ok(new
                    {
                        token = tokenString
                    });
                }

                return Unauthorized("Username atau password salah");
            }
        }
    }
}

```

Tambahkan [Authorize] di atas class

```

[Authorize]
[ApiController]
[Route("api/[controller]")]
1 reference
public class MahasiswaController : ControllerBase
{

```

5. Jalankan Program bisa menggunakan Swagger
6. Gunakan username: admin dan password: admin123, kemudian pilih Authorization, pilih type Bearer Token kemudian paste token:


```

{
  "username": "admin",
  "password": "admin123"
}

```
7. Kemudian klik send